Tic-Tac-Toe: A Python Programming Project

A Mini Project Presentation presented to the

College of Information Technology Education

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Introduction

DESCRIPTION

The Tic-Tac-Toe Game Using Python is a simple, interactive game designed for two players. The players take turns placing their symbols, X and O, on a 3×3 game board. The main goal is to get three matching symbols in a horizontal, vertical, or diagonal line to win the game. If all spaces on the board are filled and neither player wins, the game ends in a draw. This project was developed using the Python programming language and basic programming concepts such as functions, loops, conditional statements, and input validation. It aims to demonstrate how Python can be used to create a functional game while improving logical thinking and problem-solving skills.

GENERAL OBJECTIVE

To develop a Tic-Tac-Toe Game Using Python.

SPECIFIC OBJECTIVES

1. To develop a functional Tic-Tac-Toe game using Python programming language and basic programming concepts.

2. To apply Python functions and conditional statements to manage player moves, validate inputs, and determine the winner or a draw.

3. To enhance logical thinking and problem-solving skills by creating a simple, interactive game that

allows two players to compete against each other.

SCOPE AND LIMITATIONS

SCOPE

The Tic-Tac-Toe Game is a Python-based program designed for two players who take turns placing X and O on a 3×3 game board. The program displays the game board, accepts player input, checks whether moves are valid, and determines whether a player wins or the game ends in a draw. It also allows players to start a new game after finishing the current one. The program demonstrates basic Python concepts, including functions, loops, conditional statements, and input validation.

LIMITATIONS

The program is limited to two human players and does not include a computer opponent or artificial intelligence. It uses a simple text-based interface rather than a graphical user interface. The game can only be played in an environment that supports Python, such as a Python IDE or terminal. Additionally, it does not include online multiplayer, score tracking across multiple sessions, or a database for saving game records.

DEFINITION OF TERMS

Python – A programming language used to develop the Tic-Tac-Toe game.

Tic-Tac-Toe – A game in which two players compete to form a line of three matching symbols on a 3×3 board.

Function – A reusable block of code that performs a specific task in a program.

Game Board – The 3×3 grid where players place their symbols.

Input Validation – The process of checking whether a player's move is valid before accepting it.

Conditional Statement – A programming statement that allows the program to make decisions based on specific conditions.

Loop – A programming structure that repeats instructions until a condition is met.

Winner Detection – The process of checking whether a player has formed three matching symbols in a row, column, or diagonal.

Draw – A game result in which all board spaces are occupied and neither player wins.

FLOW OF THE DATA

The program starts by initializing the game board and assigning X as the first player. It displays the board and asks the current player to enter a move. The program validates the input and checks whether the selected cell is empty. If the move is invalid, the player is asked to try again. If valid, the symbol is placed on the board, and the program checks for a winner or a draw. If the game continues, the turn switches to the other player. When the game ends, the result is displayed, and the players may choose to play again or end the program.

KEY FEATURES

3×3 Game Board – Displays the available spaces for player moves. Two-Player Mode – Allows two players to take turns using X and O. Input Validation – Prevents invalid moves and occupied-cell selections. Winner Detection – Checks rows, columns, and diagonals for a winning combination. Draw Detection – Identifies when the board is full without a winner. Play Again Option – Allows players to start a new game after a match. Function-Based Structure – Organizes the program into functions for readability and easier maintenance..

SOURCE CODE

```
def display_board(board):
    """Display the Tic-Tac-Toe board."""
```

```
    print()
    print(" " + board[0] + " | " + board[1] + " | " + board[2])
    print("---+---+---")
    print(" " + board[3] + " | " + board[4] + " | " + board[5])
    print("---+---+---")
    print(" " + board[6] + " | " + board[7] + " | " + board[8])
    print()


def check_winner(board, player):
    """Check if a player has three symbols in a row."""
    winning_combinations = [
        (0, 1, 2),
        (3, 4, 5),
        (6, 7, 8),
        (0, 3, 6),
        (1, 4, 7),
        (2, 5, 8),
        (0, 4, 8),
        (2, 4, 6)
    ]

    for combination in winning_combinations:
        if all(board[position] == player
               for position in combination):
            return True
```

```
    return False



def check_draw(board):
    """Check if the board is full."""
    return all(position in ["X", "O"] for position in board)



def get_move(board, player):
    """Ask the current player to choose a position."""
    while True:
        try:
            move = int(input(
                f"Player {player}, choose a position (1-9): "
            ))

            if move < 1 or move > 9:
                print("Invalid position! Choose from 1 to 9.")
            elif board[move - 1] in ["X", "O"]:
                print("That position is already occupied!")
            else:
                return move - 1

        except ValueError:
            print("Invalid input! Please enter a number.")
```

```
def play_game():
    """Run the Tic-Tac-Toe game."""
    board = [str(number) for number in range(1, 10)]
    current_player = "X"

    print("================================")
    print("       TIC-TAC-TOE GAME")
    print("================================")
    print("Choose a position from 1 to 9.")
    print("The first player to get 3 in a row wins!")

    while True:
        display_board(board)

        move = get_move(board, current_player)
        board[move] = current_player

        if check_winner(board, current_player):
            display_board(board)
            print(f"Congratulations! Player {current_player} wins!")
            break

        if check_draw(board):
            display_board(board)
            print("It's a draw! Good game!")
            break
```

```
        # Switch to the other player
        if current_player == "X":
            current_player = "O"
        else:
            current_player = "X"



def main():
    """Main function of the program."""
    while True:
        play_game()

        again = input("Do you want to play again? (yes/no): ")
        if again.lower() != "yes":
            print("Thank you for playing Tic-Tac-Toe!")
            break



# Start the program
if __name__ == "__main__":
    main()
```